

Finite-Horizon MDP

Goal: The learning agent aims to find a policy π that maximizes the expected cumulative reward over a finite horizon

$$\pi^* \in \arg \max_{\pi \in \Pi} \mathbb{E}_{\substack{s_0 \sim \mathcal{M}_0 \\ A_t \sim \pi(z_t) \\ s_{t+1} \sim P(\cdot | s_t, A_t)}} \left[\overbrace{\sum_{t=0}^{T-1} R(s_t, a_t)}^{\text{return}} \right]$$

$\gamma^t \rightarrow + R(s_T)$
in some settings

Value functions

- (state) value function of a policy π at time $t \in \{0, 1, \dots, T-1\}$,

$V_t^\pi: S \rightarrow \mathbb{R}$, is defined as

$$V_t^\pi(s) = \mathbb{E}_{\substack{A_t \sim \pi(z_t) \\ s_{t+1} \sim P(\cdot | s_t, A_t)}} \left[\sum_{t'=t}^{T-1} R(s_{t'}, A_{t'}) \mid s_t = s \right] \quad \forall s \in S$$

$\gamma^{t'-t}$

- (state-) action value function of a policy π at time $t \in \{0, 1, \dots, T-1\}$,

$Q_t^\pi: S \times A \rightarrow \mathbb{R}$, is defined as

$$Q_t^\pi(s, a) = \mathbb{E}_{\substack{A_t \sim \pi(z_t) \\ s_{t+1} \sim P(\cdot | s_t, A_t)}} \left[\sum_{t'=t}^{T-1} R(s_{t'}, A_{t'}) \mid s_t = s, A_t = a \right] \quad \forall s \in S, \forall a \in A$$

$\gamma^{t'-t}$

Theorem: There exists an optimal, deterministic, history-independent, time-varying policy π^* , i.e.,

$$\exists \pi^* = (\pi_0^*, \pi_1^*, \dots, \pi_{T-1}^*), \quad \pi_t^* : S \rightarrow A$$

$$\text{s.t.} \quad V_t^{\pi^*}(s) \geq V_t^{\pi}(s) \quad \forall s \in S, \forall \pi \in \Pi, \forall t \in \{0, 1, \dots, T\}$$

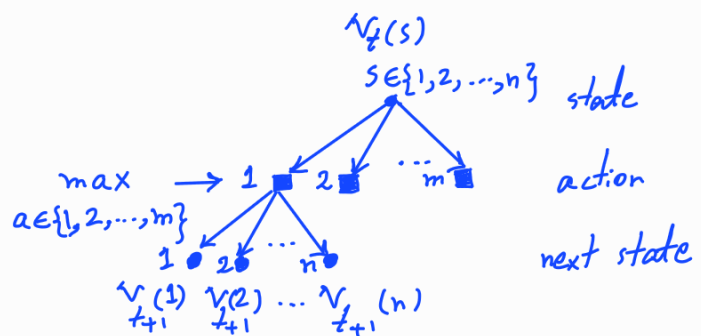
→ we look for finding an optimal policy $\pi^* = (\pi_0^*, \pi_1^*, \dots, \pi_{T-1}^*)$, which is a sequence of deterministic policies dependent on state and time.

* Bellman (consistency) Equations:

$$V_t^{\pi}(s) = \mathbb{E}_{a \sim \pi_t(s)} \left[R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [V_{t+1}^{\pi}(s')] \right]$$

$$Q_t^{\pi}(s, a) = R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [V_{t+1}^{\pi}(s')]$$

* Bellman Optimality Equations:



$$V_t^* = \max_{a \in A} \left[R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [V_{t+1}^*(s')] \right]$$

$$Q_t^*(s, a) = R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \left[\max_{a' \in A} Q_{t+1}^*(s', a') \right]$$

Note: The optimal value function is also a sequence, i.e.,

$$V^* = (V_0^*, V_1^*, \dots, V_T^*)$$

objective $\rightarrow$ V_0^* $\rightarrow$ 0

Value Iteration (for finite-Horizon MDP)

VI aims to find the (exact) optimal value function and an optimal policy through dynamic programming (backward induction).

Iterative method:

- Initialize $V_T^* = \mathbf{0}_{|S| \times 1}$

- For $t = T-1, T-2, \dots, 0$:

$$V_t^*(s) = \max_{a \in A} [R_t(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [V_{t+1}^*(s')]] \quad \forall s \in S$$

$$\pi_t^*(s) = \arg \max_{a \in A} [R_t(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [V_{t+1}^*(s')]] \quad \forall s \in S$$

- Return $V^* = (V_0^*, V_1^*, \dots, V_T^*)$ $\rightarrow$ if time-varying

and $\pi^* = (\pi_0^*, \pi_1^*, \dots, \pi_{T-1}^*)$

Note: These expressions and the VI solution for finite-horizon MDP can be extended to the case of time-varying MDPs where the transition P_t and reward R_t are time-dependent.

Effective Horizon

Question: Can we approximate an infinite-horizon discounted return with a finite-horizon (truncated) return?

$$G = \underbrace{\sum_{t=0}^{\infty} \gamma^t R(s_t, A_t)}_{\text{true return } G} = \underbrace{\sum_{t=0}^{T-1} \gamma^t R(s_t, A_t)}_{\text{truncated return } \hat{G}} + \underbrace{\sum_{t=T}^{\infty} \gamma^t R(s_t, A_t)}_{\text{error } G - \hat{G}}$$

If $|R(s, a)| \leq R_{\max} \quad \forall s \in \mathcal{S}, \forall a \in \mathcal{A}$

$$\leq \gamma^T \sum_{t=T}^{\infty} \gamma^{t-T} R_{\max}$$

$$\longrightarrow |\text{error}| = |G - \hat{G}| \leq \frac{\gamma^T R_{\max}}{1 - \gamma} \leq \epsilon$$

$$\longrightarrow T \geq \frac{\log \frac{R_{\max}}{\epsilon(1-\gamma)}}{\log \frac{1}{\gamma}} \leftarrow \geq 1-\gamma$$

for simplicity $T_{\gamma, \epsilon} := \frac{\log \frac{R_{\max}}{\epsilon(1-\gamma)}}{1 - \gamma}$

effective horizon

Note: The notion of effective horizon can be used to create approximations between infinite-horizon discounted MDPs and finite-horizon MDPs.

Episodic MDP

In the episodic setting, the interactions happen in episodes of finite but potentially varying length. The interactions start and continue similar to the infinite-horizon and finite-horizon settings, except that they end once a terminal state is reached and then restart according to the initial distribution.

$s_{\text{terminal}} \in \mathcal{S}$

Goal: The learning agent aims to find a policy π that maximizes the expected cumulative reward over an episode

$$\pi^* \in \arg \max_{\pi \in \Pi} \mathbb{E}_{\substack{s_0 \sim \mathcal{M}_0 \\ A_t \sim \pi(\gamma_t) \\ s_{t+1} \sim P(\cdot | s_t, A_t)}} \left[\overbrace{\sum_{t=0}^{\tau-1} R(s_t, a_t)}^{\text{return}} \right]$$

γ_t

Note: It is possible to create approximations between episodic MDPs and finite-horizon MDPs.

